Turning Flight II

Consider the following turbojet aircraft:

Aircraft mass	m	2000 kg
K-factor	k	0,1
Reference wing area	S_{ref}	20 m^2
Zero-lift drag coefficient	C_{D0}	0,023
Maximum lift coefficient	$C_{L,max}$	1
Maximum load factor	$n_{z,max}$	4
Thrust density dependency exponent	n_ρ	0.75
Maximum thrust at MSL	$T_{max,S}$	$0.55 \cdot m \cdot g$

a) What is the minimum turn radius (i.e. the turn radius where $C_{L,max}$ and $n_{z,max}$ are not exceeded) and the corresponding aircraft speed? Assume a height h = 0m.

b) Can this turn radius be achieved with the available propulsive power? How high is the required thrust? Assume a height h = 0m.

c) Compute the height h starting from which the available thrust is limiting the achievable turn radius? What is the minimum turn radius at this altitude? Use the standard atmosphere.

Additional data:

Density at MSL according to ISA standard atmosphere:	$\rho_{s,MSL} = 1.225 \frac{kg}{m^3}$
Pressure at MSL according to ISA standard atmosphere:	$p_{s,MSL} = 1.01325e5\ Pa$
Temperature at MSL according to ISA standard atmosphere:	$T_{s,MSL} = 288.15\ K$
Gravitational acceleration:	$g_0 = 9.80665 \frac{m}{s^2}$
Specific gas constant	$R = 287.05 \frac{J}{kg\ K}$
Polytropic exponent:	$n = 1.235$
Isentropic exponent:	$\kappa = 1{,}4$
Radius of the earth:	$r_{earth} = 6356766$ m